Genuine Multipartite Entanglement Analysis

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Model                              Ground-state properties           GME summary
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Hubbard ring (1-D, PBC)            E0                 : -4.60352630  Extracted n-spin RDM : n = 4
                                   E1                 : -4.29999276
System parameters                                                    Purity:
-----------------                  <S2>               : -0.000000      gamma = Tr(rho^2)
Sites (L)          : 8             S                  : -0.0000
Electrons (Ne)     : 8             <Sz>               : 0.000000    Separable cut criterion:
Hopping (t)        : 1.0           Multiplicity 2S+1  : 1.0           gamma_A = 1  <->  rho = rho_A x rho_B
Interaction (U)    : 4.0           Spin state         : Singlet (S=0)
Mode               : uniform                                       C_GME (Troiani et al.):
n-body order       : 4             Double occupancy                  C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))
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                                   <D> total          : 0.759406    Convex roof (mixed):
                                   <D> / site         : 0.094926      C_GME(ρ) = min Σ pᵢ C_GME(|ψᵢ⟩)


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Summary: All 4-Body RDM Clusters

System: L=8, Ne=8, U=4.0, t=1.0 | E_0=-4.603526

Sites	Purity	C_GME	C_LB	C_cayl	C_expo	C_UB	Status

Legend:
  Purity = Tr(rho²) [1.0=pure, <1.0=mixed] | C_GME = Genuine multipartite entanglement (pure only)
  C_LB = Lower bound (notes eqs.6-9) | C_cayl = unitary optimization - Cayley Transform |C_expo= unitary optimization - Expoenetial Map | BEST
  C_UB = Upper bound (eigendecomp) | Status: GME =has GME | NO GME=separable | MIXED=mixed state